Cheng-Hao (Howard), Kao

About me

A second-year MASc student in Mechatronics engineering at the University of Waterloo. Proficient in machine learning and data analysis. Possess the analytical abilities required for research and design as well as a comprehensive grasp of engineering principles. Currently researching on multi-agent system workloads as an intern at Huawei, and working in the Structural Biomechanics Lab on ML-based injury prediction. Previously involved in the Robotics and Computer Vision Lab conducting research on multi-robot collaboration joint trajectory planning and control systems and in the Microelectromechanical Systems Lab dedicated to integration of FPCB mirror-based scanners. Seeking more professional training for further development within the field of AI, data analysis, and cognitive science.

Contact

Born on 1999/11/30, Age 25

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Cheng-Hao Kao

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Languages

Chinese - Native Language

English - Professional Knowledge

Education

January 2024 - Present

MASc in Mechanical and Mechatronics Engineering — Waterloo, On, Canada
University of Waterloo
GPA: 3.90/4.0 Cumulative

Relevant Coursework: Introduction to Machine Learning, Pattern Recognition and Machine Learning, Deployment of Deep Learning Model, Computational Intelligence, Optimization Methods, Human Factors in Testing

August 2018 - June 2023

Bachelor of Engineering — Toronto, On, Canada
Toronto Metropolitan University
GPA: 3.85/4.33 Cumulative

Relevant Coursework: Mechanical Design, Control of Robotic Manipulators, Real-Time Computer Control Systems, Fundamentals of Robotics, Mechatronics Systems Design, Measurements, Sensors and Instruments, Microprocessor Systems, Control Systems Digital Systems, Computer Programming Fundamentals, Electric Machines and Actuators, Electric Circuits, Applied Finite Elements, Vibrations, Machine Design, Mechanics of Machines, Solid Mechanics, Statics and Mechanics of Materials, Dynamics, Introduction to Engineering Design, Engineering Graphical Communication, Numerical Analysis, Linear Algebra, Differential Equations and Vector Calculus, Calculus II, Calculus I

August 2015 - June 2018

High School Diploma — Shanghai, China
No.2 High School of East China Normal University
Average: 95.3/100

Work Experience

August 2025 - Present

Technology Planning and Cooperation Research Intern — Waterloo, On, Canada
Huawei Canada

- Multi-agent system analysis
- Agentic science analysis
- AI4Research analysis and development

May 2024 - April 2024

Teaching Assistant — Waterloo, On, Canada
University of Waterloo

- Teaching Circuits
- Teaching Mechanics of Deformable Solids
- Teaching Statics

June 2023 - January 2024

Research Assistant — Toronto, On, Canada
Toronto Metropolitan University RCVL Lab

- Research on multi-robot collaboration joint trajectory planning and control
- Research on micro-plastic detection via ultrasonic sensor and machine learning

Professional Skills

Deep Learning · Machine Learning · Data Analysis · Optimization · Signal Processing · Robotics · Control Systems · Microcontrollers · Simulation and Design · Engineering Mathematics

Honors and Awards

Dean's List 2018 to 2019 · Dean's List 2019 to 2020 · Dean's List 2020 to 2021 · Dean's List 2021 to 2022 · Dean's List 2022 to 2023 · Mechatronics Engineering Alumni Award · Mechatronics Engineering Capstone Project Award · Graduation with Distinction

Soft Skills and Strengths

Adaptability · Problem Solving · Collaboration · Team Working · Open-Mindedness · Critical Thinking · Communication · Willingness to Learn · Responsibility · Reliability · Enthusiasm · Autonomy · Motivated · Stress Tolerance · Research and Analysing · Personal Time Management · Strategic Planning · Patience · Empathy · Discipline

Interests

- Deep Learning
- Machine Learning
- Cognitive Science
- Agentic Systems
- Control Systems
- Robotics
- Auto Chess
- Cats
- Bees
- Board Games
- Billiards
- Squash

September 2022 - February 2023

Research Assistant — Toronto, On, Canada
Toronto Metropolitan University MEMS Lab

- Research on FPCB mirror-based lidar scanners
- Integration of PSD sensors, ToF sensors, and GUI

April 2021 - September 2021

Research Assistant Intern — Toronto, On, Canada
Toronto Metropolitan University Ultrashort Laser Nanomanufacturing Lab

- Conducting porosity and fiber analyses via SEM image processing
- Generating molecular dynamics simulations

April 2020 - September 2020

Research Assistant Intern — Toronto, On, Canada
Toronto Metropolitan University Ultrashort Laser Nanomanufacturing Lab

- Conducting porosity and fiber analyses via SEM image processing
- Generating molecular dynamics simulations

Publications

Manuscript 2025 — **Predicting the Risk of ACL Injuries Using Machine Learning Models and Tibial Anatomical Predictors**, *Accepted & Unpublished*, Link Unavailable

Arxiv Preprint 2024 — **Morphological Detection and Classification of Microplastics and Nanoplastics Emerged from Consumer Products by Deep Learning**, *IEEE*, Link

Journal Article 2023 — **MR. CAP: Multi-Robot Joint Control and Planning for Object Transport**, *IEEE Control System Letters*, Link

Other Projects

- Time-series signal classification for tool and firearm use intervention via deep learning (Honda, GRYP Tech & UW)
- Data cleaning of wearable sensor time-series data (Honda, GRYP Tech & UW)
- IMU data processing and visualization (GRYP Tech & UW)
- Object segmentation and identification via SAM and YOLO (UW)
- Analyzing effect of homogeneous bagging on small medical datasets (UW)
- EKF design for 3D object tracking (UW)
- Improving SVM prediction accuracy on minority classes for imbalanced datasets (UW)
- MPC implementation for path planning and controls (TMU)
- Inverted pendulum control via PID controllers (TMU)

Software Skills

Data Analysis	**PyTorch**	**TensorFlow**
	AutoGluon	**TabNet**
	SkLearn	**YOLO**
	SAM	**CATBoost**
	XGBoost	**Optuna**
	NLopt	
Robotics	**GTSAM**	
Modeling and Simulation	**Simulink**	**MATLAB**
	ANSYS	**Multisim**